

Quantitative Modeling of Systems in *Drosophila* Sensory Biology

Background and Context

Systems in nature - whether they be sub-cellular dynamics, or high-level interactions that constitute ecosystems - are hallmarked by their inconceivable complexity. In the endeavor of understanding such systems, it is easy (if not guaranteed) for one to become overwhelmed by the number of components and their contributions to the high-level systematic behavior. Quantitative models, or mathematical and computational representations of these systems and their various components, are capable of tracking all of these “moving parts” and their respective contributions to overall behavior. Often, biological models are built to reflect the changing states of different system components. This can be accomplished through depictions such as differential equations that relate functions representative of components and their derivatives, or a measure of their general behavior (Mogilner *et al.* 2006). In this sense, quantitative models are incredibly powerful tools that can help researchers handle these large systems and from them, discern shapes of potential system outcomes, identify major players within the system, and determine the boundaries of system behavior as a whole.

The application of such quantitative models to biological studies requires considerable experimental data on which to base the mathematical and computational representations of the components of such systems. Experimental biology research that focuses on systems at the sub-organism level often acquires data through the use of model organisms. These are live organisms employed to reflect aspects of human biology in biological research in order to circumvent ethical and monetary obstacles of such research. Additionally, the use of model

organisms allow researchers to perform experimental work at large scales (Griffiths 2010). One such model organism is *Drosophila melanogaster*, commonly known as the fruit fly. Due to the conservation of approximately 75% of known human disease genes in the species (Ajjuri *et al.* 2015), their relative simplicity and inexpensiveness, *Drosophila* make for very popular model organisms. Similarities between *Drosophila* and human biology allow for researchers to speculate on human biology given findings of experimental studies. This allows quantitative modeling of biological systems, specifically sensory systems within *Drosophila*, which can provide insights into how stimuli is captured by organisms, encoded in the brain, and then decoded into behavioral responses.

One sensory modality under study in *Drosophila* is olfaction, also known as odor perception (Farooqui 2016). *Drosophila* capture olfactory stimuli via receptors that line the olfactory epithelium on their antennae. Olfactory receptor neurons (ORNs) then transduce olfactory signals to the olfactory bulb, the brain structure that processes odor. Unique ORN types are believed to be specific for unique olfactory receptors (Couto *et al.* 2005). Specifically, these ORNs first carry stimuli information to the antennal lobe (AL), the portion of the olfactory bulb that receives initial olfactory stimuli information (Badel *et al.* 2016). The AL consists of projection neurons (PNs), or neurons that are integral in encoding and decoding of stimuli. These PNs are organized into glomeruli, or structures where synapses form between peripheral and central sensory nerves (Kosaka *et al.* 1998). Similar to ORNs, glomeruli exhibit unique activation patterns for unique stimuli, therefore quantification of these activation patterns could determine odor identities. This mapping of odor perception, coupled with difficulties in

observing mechanisms of neuronal activity, lends the mechanisms of olfactory sensory detection, encoding and decoding to potentially be unveiled via quantitative modeling.

Another sensory modality of interest in *Drosophila* is vision. At a high level, mechanisms and anatomy of sight in *Drosophila* are more of a departure from the link between humans and flies in senses such as olfaction. While flies may not seem to be quite as strong of a “model” for vision research, fly sight still provides important insights into the mechanisms of vision at smaller biological scales. *Drosophila* possess a compound eye, an eye comprised of multiple visual units. This has a fundamental effect on the organisms’ visual processing (Paulk *et al.* 2013). There are approximately 750 of these units, or ommatidia, per eye. Each ommatidium contains all eight types of photoreceptor cells, which convert light into biological signals and then transduce these signals to the optic lobe for processing (Hardie 1985). Vision is essential in *Drosophila* for phototaxis, or navigation in response to visual stimuli, as well as mate recognition. However, the neural mechanisms behind visual information integration such as spatial memory and pattern recognition are poorly understood (Dewar *et al.* 2017). This problem lends itself to be examined by computer vision methods.

Other types of quantitative models have also been proposed to unveil mechanisms of vision such as motion detection. The Hassenstein-Reichart detector is an early model that captures how *Drosophila* detect motion and its direction by determining the changes in luminance, or the directional intensity of light over a certain area, across two adjacent visual units over a certain amount of time. At the time of model development, the mechanisms of this process remained unclear. However, nearly 50 years later headway was made on the determination of a biological basis for this process of motion detection by linking two cells types

that make up the connection of the two visual units (Yonehara and Roska 2013). This phenomenon of theoretical findings later showing up or becoming useful in research once sufficient technology has been established has appeared across disciplines including mathematics and its applications in computer science, and shows the power that theoretical and modeling approaches have in biological research.

Chemotaxis, the multi-sensory based process of navigation, is intimately connected to olfaction and vision in *Drosophila*. The mechanisms underlying the assimilation of these different sensory stimuli, how conflicting stimuli are prioritized or reconciled, and how chemotactic decisions are made is also poorly understood (Gepner *et al.* 2015). Due to the various sensory components that contribute to chemotaxis, the process lends itself to being modeled in a quantitative manner.

State of the Field

A high-level, general structure of models that have been proposed for the pathways of sensory information can be split into two objectives: first, the linking of stimuli and neural codes, and second, the linking of neural codes and behavior, in order to ultimately bridge these and determine the correlation between stimuli and behavior (Clemens and Murthy 2017). These models often initially treat the brain as a “black box”, or a system that can be considered only in terms of its inputs and outputs in order to simplify its inner workings as much as possible. Complexity can then be iteratively built up to determine the mechanisms behind sensory integration and response at various levels, including the cellular and sub-cellular levels of these processes that still remain significantly unclear. A most-obvious benefit of such models are that as opposed to experimental biology research, their behavior is determined by a set of rules, and

so it can be assessed in an unambiguous manner (Ingalls 2013). Thus, they can act as a construct of a working hypothesis of various sensory systems, with explicit justification of why it functions in the manner that it does, and so these models give clear insight into the potential physical structure, logic and limitations of sensory systems in *Drosophila*.

In the attempt to understand the encoding and decoding of olfaction in the *Drosophila* brain, a comprehensive model of the glomeruli in the olfactory bulb has been created (Badel *et al.* 2016). The primary goal of the model's creation was to determine whether a labeled-line theory, or the idea that individual neurons may carry sensory information from a unique receptor type and thus a unique stimulus, was a coding strategy that was employed across all glomeruli, or just a specific subset. The model takes advantage of the evidence that glomeruli act as an "odor map" in this portion of the brain and simulates glomeruli activity patterns, normalizes their activity, sums the resulting normalized activity, and outputs a valence index, or a likelihood score of a fly's preference for a given odor. In order to make the model reflect the real behavior of *Drosophila* in response to these odors, this normalization step had to be introduced to the model, providing the insight that some sort of physical normalization action occurs in the fly's brain in order to properly encode olfactory stimuli and decode this information into preferable behavior. Further, due to the model's increased confidence in the valence index output with increased numbers of glomeruli, the researchers were able to determine that the labeled line theory was likely a general coding strategy for all glomeruli in the olfactory bulb of *Drosophila*.

Models have also been proposed to help explain the mechanisms behind motion detection in *Drosophila* vision. In the attempt to understand how neurons associated with visual response function, neuronal activity data was collected and employed to build a neural network

representation of *Drosophila* vision (Dewar *et al.* 2017). This network was far more successful at identifying general information about objects' size, orientation and position than it was at more abstract pattern recognition, suggesting that *Drosophila* integrate visual stimuli in the former manner, as opposed to previous propositions of the latter.

Another model proposed to understand *Drosophila* vision attempts to capture how the organisms respond to light. The model predicts firing rates of neurons in response to light stimuli by fitting the stimuli to a linear filter that relates temporal processing to the neuronal response, and then applying a nonlinearity to the resulting fitted data in order to link the encoded stimulus and behavioral response (Baccus and Meister 2002). This model helped determine that light stimulus is handled differently depending on its contrast, as the kinetics of the filters must change depending on the contrast level in the visual context of the stimulus, and so the neurons handling the mediation of transducing such stimuli must have an adaptive capacity for such situations.

Various models have also been proposed to reveal the mechanisms and computations behind *Drosophila*'s capacity to integrate sensory information of multiple types. One primary concern within multi-sensory integration is how organisms prioritize different stimuli types, especially in conflicting scenarios. One model built to reflect this system determined that computations performed in the *Drosophila* brain can be accurately reflected by the assignment of negative values to blue light, the type of light visible to the organisms, but assign positive values to red light, light that in this experiment was used to activate odor-sensing nerve cells (Gepner *et al.* 2015). In the case of exposing an organism to both stimuli, simulating a conflicting scenario, it was shown that the organism arrives at a behavioral decision simply by summing the values

linearly according to their intensity to dictate their behavioral response. This summation was reflected in the ultimate model design that best fit the experimental data, providing insight into some incidences of relative simplicity in sensory encoding and decoding in the *Drosophila* brain.

Another focus of modeling to explain behaviors that require integrated sensory encoding and decoding are models proposed to explain chemotaxis. One such model made use of a computer vision algorithm for detecting *Drosophila* orientation in different olfactory and visual contexts in experimental procedures to obtain behavioral data (Gomez-Marin *et al.* 2011). The study also developed a Markov model, or a model used to capture stochastic, or seemingly randomly-changing systems, in order to separate the different types of *Drosophila* orientation into four separate states. By training this model on real data, transition probabilities between states were able to be determined, and numerical simulations of state transitions were able to be performed. With these computational tools, it was determined that gradients of different sensory stimuli can predict the direction of resulting *Drosophila* orientation.

Future Directions

While models of different sensory systems at various levels in *Drosophila* are being created more frequently than ever, there are numerous systems that have not yet been considered in a modeling scenario. One such focus is the ability of *Drosophila* to perform menotaxis, or navigation with respect to a light source, specifically the sun. They are able to perform this navigation type via cells of the central complex, a brain region highly conserved amongst insects, but the mechanism behind this navigational cells remains unknown (Giraldo *et al.* 2018). Computational modeling of this phenomena can help researchers reveal the mechanisms of this sensory capability.

Due to the difficulty in performing experimental neuroscience research, capturing the dynamics and mechanisms of the encoding and decoding of stimuli signals in the brain has been a central goal in modeling of sensory stimuli-behavioral response systems. Previously, imaging methods have been far too slow to capture the speed of propagation of such electrical signals. However, methods are being developed to help visualize signal propagations and activation in the brain at these critical speeds. These such methods are approaching the capacity to record whole-brain activity in *Drosophila* (Aimon *et al.*, 2015). While these methods visualize activity and cannot give precise mechanisms, pathways or reasoning behind different stimuli-response activity, this research in conjunction with research on understanding the roles and responsibilities of brain components can help to give immense insight into these activity, and can improve the efficiency and accuracy of modeling research. Moving forward, these improved imaging approaches could be integrated into the building of encoding-decoding models in order produce models with more thorough support.

In reality, an organism's sensory response behavior is often far more nuanced than a simple input-output machine. An individual organism's previous experiences can have a massive impact on the likelihood of their different reactions, but they are often unaccounted for in stimuli-response studies. In order to account for such hidden factors for decision making, statistical models have been built to estimate and correct for these "serial dependencies" that complicate probabilities of behavior in humans (Fründ *et al.* 2014). The development of this method determined that even trained test subjects are subject to such serial dependence, and so the application of this type of analysis to different organisms would certainly provide insight into how different species weigh their current environment and biological circumstances versus their

previous experiences. Perhaps the application of such methods to *Drosophila* could also provide insight into the level of stimuli experience memory that they experience, and how these factors have differential effects on different species.

These examples of future model considerations hardly capture the potential for the future of sensory models. Computational modeling, especially that of complex biological systems, is still in its infancy in terms of its comprehensiveness, power, and speed. Ultimately, these models are hoped to sometimes replace live organisms in order to address future research in a faster, more cost-effective, and ethical manner. The field of biological modeling has already provided considerable insights into various biological sensory systems that have previously presented huge obstacles due to the “hidden behavior” of the brain. However, considering the abundance of situations that have not been considered in these models, it is clear that there are certainly monumental discoveries waiting to be made with this increasingly powerful approach to biological research.

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